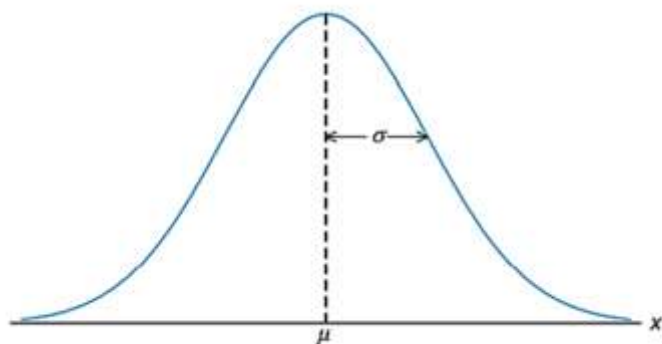


Normal Distribution

In probability theory, the normal distribution is a very common continuous probability distribution.

Many continuous variables have distributions that are bell-shaped and are called approximately normally distributed variables.

The theoretical curve, called the bell curve^{*} or the Gaussian distribution, (German Mathematician) can be used to study many variables that are not normally distributed but are approximately normal.



The density of the normal random variable X , with mean μ and variance σ^2 , is

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty,$$

where $\pi = 3.14159 \dots$ and $e = 2.71828 \dots$

A normal distribution with a mean 0 and standard deviation of 1 is called the **standard normal distribution**.

The random variable of a standard normal distribution is known as the standard score or a **z-score**. It is possible to transform every normal random variable X into a z score using the following formula:

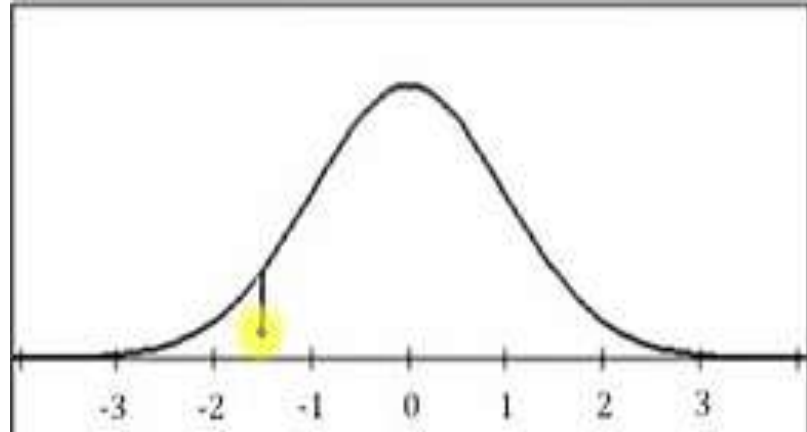
$$z = (X - \mu) / \sigma$$

Finding probabilities using a table (left of z-score)

1. The percentile in the table gives the percent less than the Z-score

2. To find the percent more than the Z-score take $100\% - \text{percent from the table}$

3. To find the percent between two Z-scores, subtract the smaller percent from the larger percent.



QUIZ # 3

Q.1

TIME :- 5 MINTS

MARKS:- 5

The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly 5 of the next 7 patients having this operation survive.

Q.1

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond

Standard Normal Probabilities

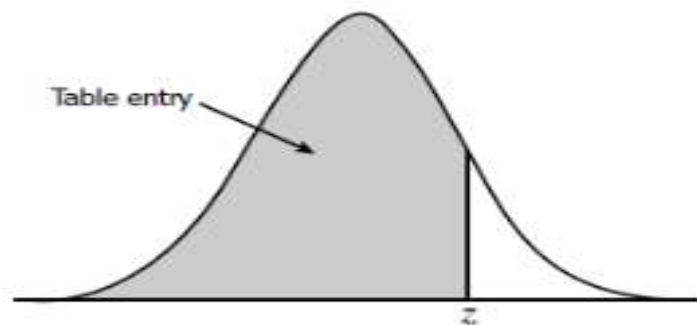


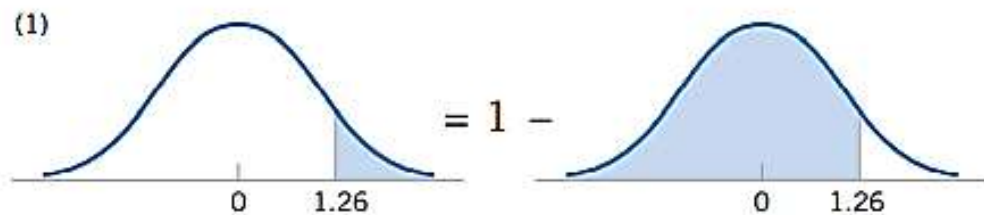
Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177

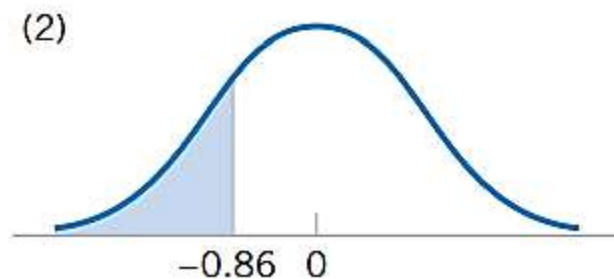
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

i

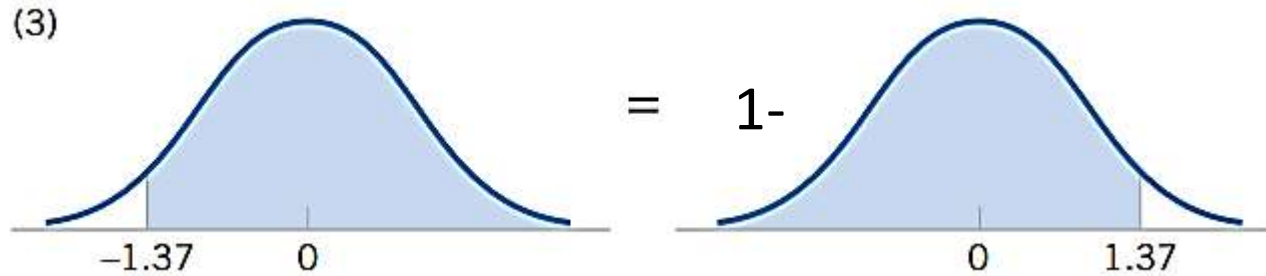
$$(1) \quad P(Z > 1.26) = 1 - P(Z \leq 1.26) = 1 - 0.89616 = 0.10384$$



$$(2) \quad P(Z < -0.86) = 0.19490.$$



3. $P(x > -1.37)$



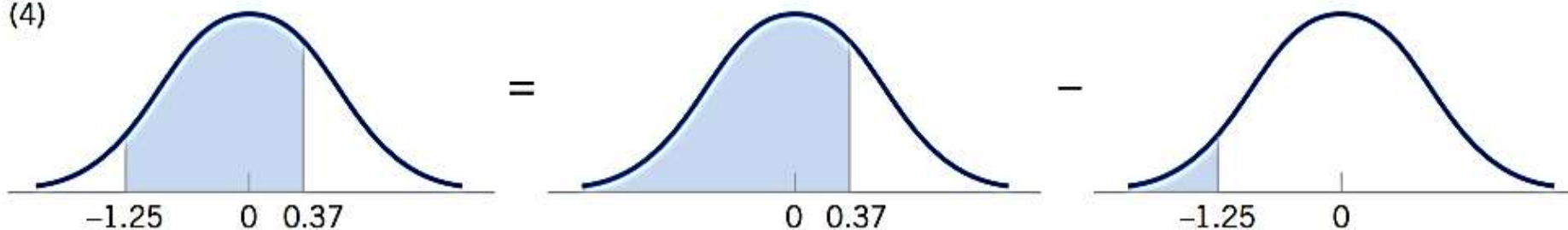
$$= 1 - P(x < -1.37)$$

$$= 1 - 0.08534$$

$$= 0.9146$$

(4) $P(-1.25 < Z < 0.37)$.

(4)



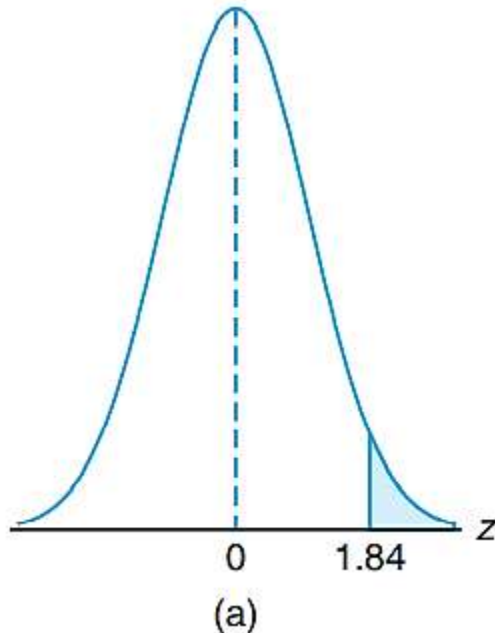
$P(Z < 0.37) - P(Z < -1.25)$. Now,

$P(Z < 0.37) = 0.64431$ and $P(Z < -1.25) = 0.10565$

$P(-1.25 < Z < 0.37) = 0.64431 - 0.10565 = 0.53866$

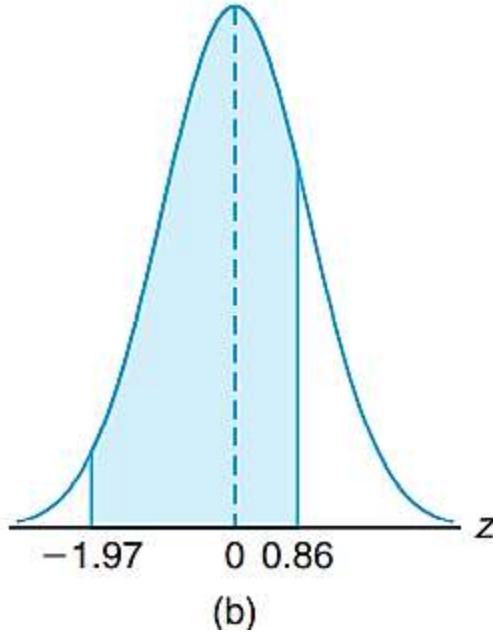
Example 6.2: Given a standard normal distribution, find the area under the curve that lies (a) to the right of $z = 1.84$ and

(a) The area in Figure (a) to the right of $z = 1.84$ is equal to 1 minus the area $1 - 0.9671 = 0.0329$.



z	0.00	0.01	0.02	0.03	0.04
0.0	0.5000	0.5040	0.5080	0.5120	0.5160
0.1	0.5398	0.5438	0.5478	0.5517	0.5557
0.2	0.5793	0.5832	0.5871	0.5910	0.5948
0.3	0.6179	0.6217	0.6255	0.6293	0.6331
0.4	0.6554	0.6591	0.6628	0.6664	0.6700
0.5	0.6915	0.6950	0.6985	0.7019	0.7054
0.6	0.7257	0.7291	0.7324	0.7357	0.7389
0.7	0.7580	0.7611	0.7642	0.7673	0.7704
0.8	0.7881	0.7910	0.7939	0.7967	0.7995
0.9	0.8159	0.8186	0.8212	0.8238	0.8264
1.0	0.8413	0.8438	0.8461	0.8485	0.8508
1.1	0.8643	0.8665	0.8686	0.8708	0.8729
1.2	0.8849	0.8869	0.8888	0.8907	0.8925
1.3	0.9032	0.9049	0.9066	0.9082	0.9099
1.4	0.9192	0.9207	0.9222	0.9236	0.9251
1.5	0.9332	0.9345	0.9357	0.9370	0.9382
1.6	0.9452	0.9463	0.9474	0.9484	0.9495
1.7	0.9554	0.9564	0.9573	0.9582	0.9591
1.8	0.9641	0.9649	0.9656	0.9664	0.9671

(b) between $z = -1.97$ and $z = 0.86$.



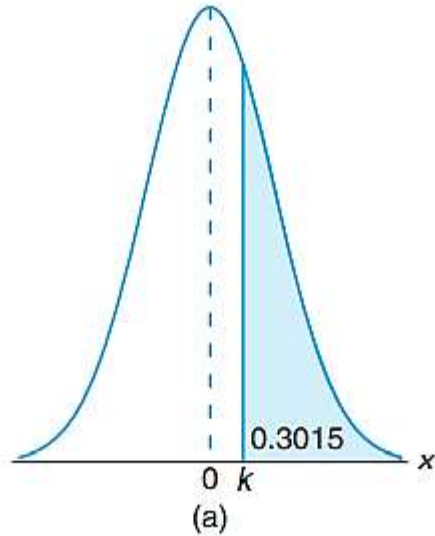
The area in (b) between $z = -1.97$ and $z = 0.86$ is equal to the area to the left of $z = 0.86$ minus the area to the left of $z = -1.97$. we find the desired area to be

$$0.8051 - 0.0244 = 0.7807$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068		
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089		
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116		
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150		
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192		
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244		

Example 6.3: Given a standard normal distribution, find the value of k such that
(a) $P(Z > k) = 0.3015$



Take complement = $1 - 0.3015$
= 0.6985
 $k = 0.52$

Property

$$P(a < Z < b) = P(Z < b) - P(Z < a).$$

(b) Find $P(k < Z < -0.18) = 0.4197$.

Solution

$$P(Z < -0.18) - P(Z < k) = 0.4286 - P(Z < k) = 0.4197 \Rightarrow k = -2.37$$

Example 6.4: Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

$$x_1 = 45 \text{ and } x = 62 \xrightarrow{\text{transformation}} z_1 = \frac{45 - 50}{10} = -0.5, \quad z_2 = \frac{62 - 50}{10} = 1.2$$

$$\begin{aligned} P(45 < X < 62) &= P(-0.5 < Z < 1.2) \\ &= P(Z < 1.2) - P(Z < -0.5) = 0.8849 - 0.3085 = 0.5764 \end{aligned}$$

Example 6.5: Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.

$$z = \frac{362 - 300}{50} = 1.24.$$

Hence,

$$P(X > 362) = P(Z > 1.24) = 1 - P(Z < 1.24) = 1 - 0.8925 = 0.1075.$$

Example:-

Suppose the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 mill amperes and a variance of 4 (*mill amperes*)². What is the probability that a measurement will exceed 13 milliamp.

Solution :-

$$P(X > 13).$$

$$\text{Let } Z = (X - 10)/2.$$

$$P(X > 13) = P(Z > 1.5) = 1 - P(Z \leq 1.5) = 1 - 0.93319 = 0.06681$$

b. what is the probability that a current measurement is between 9 and 11 mill amperes?

Solution:-

$$P(9 < X < 11) = P\left(\frac{9 - 10}{2}\right) < Z < P\left(\frac{11 - 10}{2}\right)$$

$$\begin{aligned} &= P(-0.5 < Z < 0.5) = P(Z < 0.5) - P(Z < -0.5) \\ &= 0.69146 - 0.30854 = 0.38292 \end{aligned}$$

Example

The diameter of a shaft in an optical storage drive is normally distributed with mean 0.2508 inch and standard deviation 0.0005 inch. The specifications on the shaft are 0.2500 ± 0.0015 inch. What proportion of shafts conforms to specifications?

$$\begin{aligned} P(0.2485 < X < 0.2515) &= P\left(\frac{0.2485 - 0.2508}{0.0005} < Z < \frac{0.2515 - 0.2508}{0.0005}\right) \\ &= P(-4.6 < Z < 1.4) = P(Z < 1.4) - P(Z < -4.6) \\ &= 0.91924 - 0.0000 = 0.91924 \end{aligned}$$

Example 6.6: Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

- (a) 45% of the area to the left and
- (b) 14% of the area to the right.

$$\mu = 40$$
$$\sigma = 6$$

$$x = 39.22$$

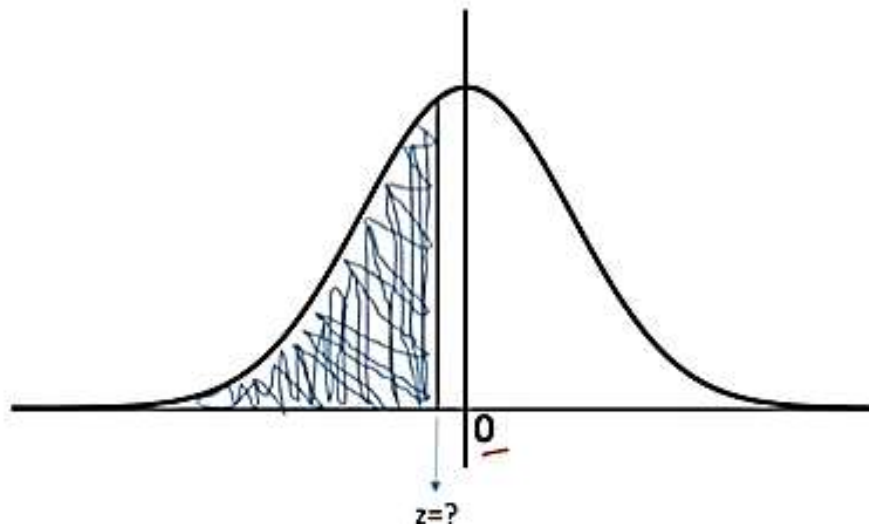
$$P(z < ?) = 0.45$$

$$P(z < -0.13) = 0.45$$

$$z = \frac{x - \mu}{\sigma}$$

$$-0.13 = \frac{x - 40}{6}$$

$$-0.13 \times 6 + 40 = x$$



Example 6.6: Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has
(a) 45% of the area to the left and
(b) 14% of the area to the right.

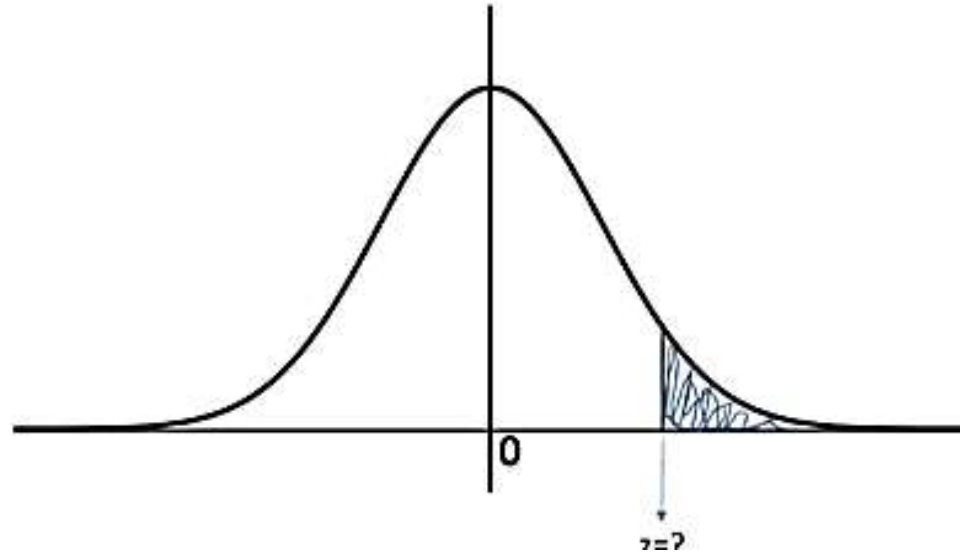
$$\mu = 40$$
$$\sigma = 6$$

$$P(Z > ?) = 0.14$$

**Take compliment = $1 - 0.14$
= 0.86**

$$P(Z < ?) = 0.86$$

$$P(Z < 1.08) = 0.86 \text{ from table}$$



$$z = \frac{x - \mu}{\sigma}$$

$$1.08 = \frac{x - 40}{6}$$

$$1.08 \times 6 + 40 = x$$

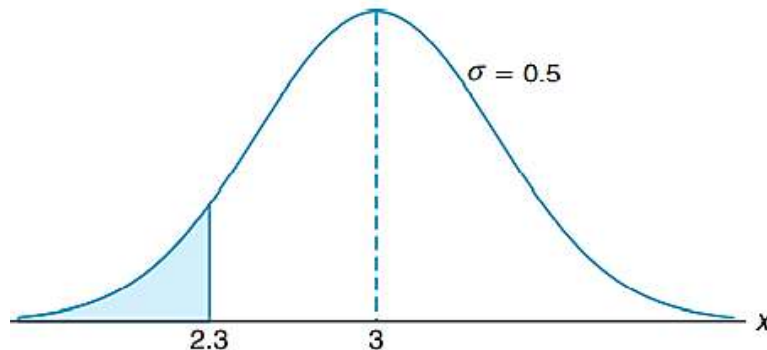
$$x = 46.48$$

Example 6.7:

A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

• Solution:

$$z = \frac{2.3 - 3}{0.5} = -1.4 \Rightarrow P(X < 2.3) = P(Z < -1.4) = 0.0808$$



Example 6.8:

An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours

Solution:. The z values corresponding to $x_1 = 778$ and $x_2 = 834$

$$z_1 = \frac{778 - 800}{40} = -0.55 \text{ and } z_2 = \frac{834 - 800}{40} = 0.85.$$

Hence,

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55) \\ &= 0.8023 - 0.2912 = 0.5111. \end{aligned}$$

EXAMPLE

The line width of for semiconductor manufacturing is assumed to be normally distributed with a mean of 0.5 micrometer and a standard deviation of 0.05 micrometer.

- (a) What is the probability that a line width is greater than 0.62 micrometer?
- (b) What is the probability that a line width is between 0.47 and 0.63 micrometer?
- (c) The line width of 90% of samples is below what value?

$$z = \frac{(x - \text{mean})}{sd}$$

$$z = \frac{(0.62 - 0.5)}{0.05} = 2.40$$

$$P(x > 0.62) = P(Z > 2.40) = 1 - P(Z < 2.40) = 1 - 0.9918 = 0.0082$$

$$(b)P(X > 0.47 \text{ and } X < 0.63) = P(X < 0.63) - P(X < 0.47)$$

:

$$\text{z-score} = (0.63 - 0.5) / 0.05 = 2.6$$

:

$$\text{z-score} = (0.47 - 0.5) / 0.05 = -0.6$$

$$P(Z < -2.6) - P(Z < -0.6) = 0.9953 - 0.274 = 0.7211$$

:

Example

The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce.

- (a) What is the probability a fill volume is less than 12 fluid ounces?
- (b) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped?
- (c) Determine specifications that are symmetric about the mean that include 99% of all cans.

Let X be the ball volume. Find $P(X < 12.25)$
Find Z -values by using $Z = \frac{X - \mu}{\sigma}$

$$Z = \frac{12.25 - 12.4}{0.1} = -1.5$$

$$P(X < 12.25) = P(Z < -1.5) = 0.0668.$$

b) $P(X < 12.1) + P(X > 12.6) \rightarrow \text{---} \textcircled{1}$
take $X = 12.1$.

$$Z = \frac{X - \mu}{\sigma} = \frac{12.1 - 12.4}{0.1} = -3.$$

take $X = 12.6$,

$$Z = \frac{12.6 - 12.4}{0.1} = 2.$$

$\Rightarrow$

$$P(Z < -3) + P(Z > 2)$$

$$P(Z < -3) + [1 - P(Z < 2)]$$

$$0.00135 + 1 - 0.9772 = 0.0241.$$

c) we find x such that $P(X \leq x)$

$$P(X > x) = 0.10$$

$$1 - P(X \leq x) = 0.10$$

$$1 - 0.10 = P(X \leq x)$$

$$P(X \leq x) = 0.90 \quad \text{--- (1)}$$

take $X = x$.

$$Z = \frac{x - 12.4}{0.1}$$

① $\Rightarrow$

$$P\left(Z \leq \frac{x - 12.4}{0.1}\right) = 0.90$$

$$\text{Let } \frac{x - 12.4}{0.1} = 0.90$$

$$x - 12.4 = 0.90 \times 0.1$$

$$x = (0.90 \times 0.1) + 12.4$$

$$\boxed{x = 12.528}$$